When Contrastive Learning Meets a Vision Foundation Model

A Doomed Fine-Tuning Experiment and Its Profound Lessons

DINOv3 Local Feature Matching for Camera Pose Estimation
Final Technical Report

The Chinese University of Hong Kong, Shenzhen
Image Processing and Computer Vision — Course Project
April 2026

We attempted to fine-tune DINOv3—a vision foundation model pre-trained on 170 million images—using contrastive learning to improve its performance on local feature matching and camera pose estimation. Across three paradigms, two architectures, three loss functions, two datasets, and seven systematic experiments, we found that this approach is **methodologically infeasible**. The failure arises from a fundamental contradiction: DINOv3's semantic features and the geometric features required for matching are inherently different entities, and geometric distinctiveness has already been irreversibly eliminated as "noise" during pre-training. This report documents the complete experimental journey, the root-cause diagnosis spanning seven layers of analysis, and the definitive conclusion that proving *why* something cannot work holds equal scientific value to proving *what* can.

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Table 1: DINOv3 ViT-L/16 specifications.

Property	Value
Architecture	ViT-Large (24 blocks, 16 heads)
Parameters	~300M
Feature dimension	1024
Patch size	16×16 px
Pre-training data	~170M images (LVD-1689M)
Pre-training	DINO + iBOT + Register Tokens
Position encoding	2D RoPE
Regularization	LayerScale

1 Introduction

1.1 Background

Self-supervised vision models such as DINOv2 and DINOv3 have demonstrated remarkable capabilities [1], [2]: without manual annotation, they learn to understand object boundaries, semantic categories, and even 3D structure within images.

DINOv3 [2], [3] is a Vision Transformer (ViT-L/16) released by Meta AI, pre-trained on approximately 170 million images. Its self-supervised training objective yields features with strong *semantic smoothness*: patches of a "door handle" resemble other "door handle" patches; patches of a "white wall" resemble patches of the same white wall from a different viewpoint. This property gives DINOv3 non-trivial zero-shot matching capability.

1.2 Problem Statement

The problem we address is:

> **Given two photographs of the same scene taken from different viewpoints, can we automatically find pixel-level correspondences between them and recover the relative camera pose?**

This is a cornerstone problem in 3D reconstruction, visual localization, and SLAM. The evaluation pipeline proceeds as follows: (1) establish dense patch correspondences via mutual nearest neighbor (MNN) matching; (2) estimate the Essential Matrix E from correspondences using USAC [4]; (3) decompose E into relative rotation R and translation t via SVD and cheirality check [5]; (4) compute angular error between estimated and ground-truth pose; (5) report AUC@5°, AUC@10°, AUC@20° and *Precision* (fraction of matches satisfying the epipolar constraint within 5×10^{-4}).

1.3 Hypothesis (Later Falsified)

DINOv3's zero-shot features already produce usable matches (Precision ≈ 32–49%). If we use geometric correspondences as a supervisory signal to fine-tune the model with contrastive learning, matching accuracy should improve further.

1.4 Result Overview

Every fine-tuning run degraded performance. The best fine-tuned result remained 10% below the zero-shot baseline. Seven experiments, seven failures, three paradigms, zero exceptions.

2 Technical Foundation

2.1 DINOv3 Architecture

DINOv3 employs a ViT-L/16 backbone [6]. For a 448×448 input, it partitions the image into $28 \times 28 = 784$ patches, each producing a 1024-dimensional feature vector.

Table 2: Dataset characteristics.

Property	**NAVI [10]**	**ScanNet [11]**
Scene type	Single-object multi-view	Indoor scene scans
Test pairs	3000	1500
Training pairs	5596	1500 (reuses test pairs)
Resolution	Max dim 1024	640×480
Depth maps	Available (precise 3D)	Unavailable (epipolar)

Key architectural components: **2D RoPE** encodes patch (row, col) positions by rotating query and key vectors, supporting native extrapolation. **LayerScale** scales each sub-layer output by a learnable channel-wise factor initialized to $\varepsilon = 10^{-5}$. **Register Tokens** (4 learnable tokens) absorb global information, letting patch tokens focus on local structure; discarded during extraction.

2.2 Self-Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{1}$$

where $Q = XW^Q$, $K = XW^K$, $V = XW^V$.

2.3 LoRA: Low-Rank Adaptation

LoRA [7] injects trainable low-rank matrices into frozen weights:

$$h = Wx + BAx \tag{2}$$

where $W \in \mathbb{R}^{d\times d}$ is frozen, $A \in \mathbb{R}^{r\times d}$ is Kaiming-initialized, $B \in \mathbb{R}^{d\times r}$ is zero-initialized, $r \ll d$ ($r = 4$). **Critical**: $B = 0$ ensures $\Delta W = 0$ at initialization—the model initially produces *exactly* the zero-shot output. LoRA was injected into Q, K, V, and output projection layers across all 24 blocks, yielding ~0.39M trainable parameters (0.13% of the backbone).

2.4 Contrastive Learning and InfoNCE

Contrastive learning [8], [9] pulls positive pairs together and pushes negatives apart. InfoNCE is the dominant formulation:

$$\mathcal{L}_i = -\log \frac{\exp(s_i^+/\tau)}{\exp(s_i^+/\tau) + \sum_{j=1}^{K} \exp(s_j^-/\tau)} \tag{3}$$

where s_i^+ is the positive-pair cosine similarity, s_j^- are K negative similarities, and τ is the temperature.

The $\tau = 0.07$ problem: Similarities are amplified by $1/0.07 \approx 14.3\times$, causing softmax to approximate argmax—only the single most confusable negative receives meaningful gradient. When the feature space fully collapses to random noise, InfoNCE converges to its theoretical expectation:

$$\mathbb{E}[\mathcal{L}] = \ln(K + 1) \tag{4}$$

With $K = 128$ negatives: $\ln(129) \approx 4.8598$. Observing this exact value constitutes **mathematical proof of mode collapse**.

2.5 Datasets

3 Experimental Journey

We report all seven experiments chronologically. Each is described with its hypothesis, method, result, and diagnosis.

3.1 Phase 0: Zero-Shot Baseline

Method. Extract 1024-dim features from frozen DINOv3; MNN matching; USAC + 5-point pose estimation.

Results. NAVI: AUC@5 = 0.24, AUC@10 = 1.07, AUC@20 = 3.31, **Precision = 49.17%**; ScanNet: AUC@5 = 0.23, AUC@10 = 1.18, AUC@20 = 4.44, **Precision = 32.20%**.

Analysis. Semantic smoothness enables non-trivial matching without task-specific training. This is our baseline.

3.2 Phase 1: Projection Head + InfoNCE — Catastrophic Collapse

Hypothesis. A learnable projection head (1024→640→256) trained with InfoNCE can map DINO features into a more discriminative matching space.

Method. Freeze backbone; add randomly initialized projection head; train with InfoNCE ($\tau = 0.07$, $K = 128$).

Results. NAVI Precision: 49.17% → **3.16%** (↓93.6%); ScanNet Precision: 32.20% → **3.47%** (↓89.2%).

Diagnosis. The training loss converged exactly to $\ln(129) \approx 4.8598$—the theoretical random-guessing limit. This is **mathematical proof of mode collapse**: all features collapsed to a single point on the hypersphere.

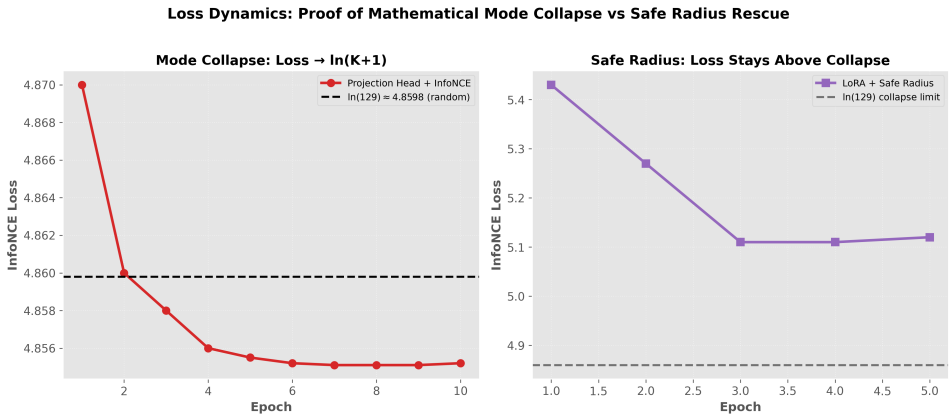

Figure 1: Left: Projection Head loss converges to ln(129), the signature of mode collapse. Right: Safe Radius keeps loss above the collapse threshold.

Lesson. Randomly initialized components destroy pre-trained features. Fine-tuning must guarantee the starting point equals zero-shot quality.

3.3 Phase 2: LoRA + InfoNCE — Zero-Init Is Not Enough

Hypothesis. LoRA with $B = 0$ initialization guarantees training starts from zero-shot quality; it should only improve or maintain performance.

Method. LoRA adapters ($r = 4$) on all QKV layers; train with InfoNCE ($\tau = 0.07$).

Results. ScanNet Precision: 32.20% → **5.51%** (↓82.9%). Model collapsed despite zero-init guarantee.

Diagnosis. InfoNCE itself is incompatible with DINOv3's feature space. It demands that *all* non-corresponding patches be pushed apart, but DINOv3's space naturally places semantically similar patches close together. The model destroys its own semantic structure to satisfy the loss.

3.4 Phase 3: Safe Radius — The Closest Brush with Success

Hypothesis. Spatially adjacent patches lie on the same surface and *should* have similar features. Excluding them from the negative set prevents destructive repulsion.

Method. Introduce a **Safe Radius** of 5 patches: negatives within spatial distance 5 from the anchor are masked out of the contrastive computation.

Results. ScanNet Precision recovered from 5.51% to **28.83%**—only 10.5% below the zero-shot 32.20%. This is the single best fine-tuned result.

Analysis. Safe Radius proved "unreasonable negatives" were a major degradation cause. Two limitations: (1) only spatial proximity is addressed, not semantic similarity (distant white-wall patches still repelled); (2) Precision recovery did not translate to AUC recovery (AUC@20 = 1.84 vs. zero-shot 4.44).

3.5 Phase 4: StableMatchingLoss — Removing the Softmax

Hypothesis. Softmax competition in InfoNCE is the root problem. A margin-based loss should be gentler.

Method. Moved to RTX 5090 (32 GB, batch_size = 8). Designed StableMatchingLoss:

$$\mathcal{L} = \underbrace{(1 - s_{\text{pos}})^2}_{\text{attraction}} + \lambda_{\text{neg}} \underbrace{\text{ReLU}(m - s_{\text{pos}} + s_{\text{hardest}})}_{\text{hardest negative only}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}} + \lambda_{\text{reg}} \underbrace{\|\mathbf{f}_{\text{ft}} - \mathbf{f}_{\text{zs}}\|^2}_{\text{feature preservation}} \tag{5}$$

Results. Worse than LoRA + Safe Radius: NAVI Precision 5.13%, ScanNet 10.97%.

Lesson. The problem is not the loss function's form, but the optimization objective itself. Improving cosine similarity bears no differentiable relationship to improving pose accuracy—the MNN, USAC, and decomposition steps are invisible during training.

All engineering variables eliminated.
The problem must be methodological.

3.6 Phase 5: Route A — Matchability Predictor

Paradigm shift. All previous methods modified DINO's feature space. Route A keeps features frozen and trains only a tiny network (~0.28M params):

$$\text{score}_i = \sigma\big(\text{MLP}(\mathbf{f}_i^{\text{DINO}})\big) \in [0, 1] \tag{6}$$

During inference, only high-scoring patches participate in MNN matching.

Results. NAVI Precision 5.55%, ScanNet 5.95%—the predictor filtered out *good* patches rather than bad ones.

Failure root cause. A single patch's feature vector does not encode how *unique* it is within the image. Two white-wall patches have nearly identical features; the predictor cannot distinguish them. This directly motivated Route B.

3.7 Phase 6: Route B — Attention Matching Head

Method. A SuperGlue-style [12] architecture with frozen DINO features: (1) DINOv3 extracts 1024-dim features; (2) linear projection 1024 → 256; (3) add 2D sinusoidal positional encoding; (4) 4 alternating blocks of Self-Attention → Cross-Attention → FFN; (5) L2 normalization; (6) **Dual-Softmax Matching**.

Total parameters: ~1M. Training objective: Dual-Softmax NLL Loss. On 784×784 score matrices, the random baseline is $2 \times \ln(784) \approx 13.33$.

Results: All metrics exactly 0.00%. NAVI: Training loss 13.33 → 12.15 (8.9% reduction; correct-match probability from 1.6×10^{-6} to 5.2×10^{-6}—still random under argmax). ScanNet: Training loss 13.33 → 13.33 (**zero learning**). This is the most complete failure.

Three-layer diagnosis:

1. **Extreme label sparsity.** Of $784^2 = 614{,}656$ possible pairs, only ~150 have labels (0.02%). The remaining 614,500 positions drift unconstrained.
2. **Zero geometric distinctiveness.** Fifty white-wall patches occupy nearly identical positions in semantic space. Their attention scores are nearly equal, producing uniform attention distributions with zero discriminative power.
3. **Data starvation.** 1M-parameter attention model trained on 5,596 pairs (NAVI) and 1,500 pairs (ScanNet). ScanNet's frozen loss (13.33→13.33) directly proves insufficient data.

4 Complete Results

Table 3: All seven experiments: raw results.

#	Method	Dataset	AUC@5	AUC@10	AUC@20	Precision
0	Zero-Shot (baseline)	NAVI	0.24	1.07	3.31	49.17%
		ScanNet	0.23	1.18	4.44	32.20%
1	Projection Head + InfoNCE	NAVI	0.00	0.03	0.10	3.16%
		ScanNet	0.00	0.07	0.25	3.47%
2	LoRA + InfoNCE (no SafeR)	ScanNet	0.00	0.09	0.96	5.51%
3	**LoRA + Safe Radius (bs=1)**	**ScanNet**	0.00	0.26	1.84	**28.83%**
4	LoRA + StableMatchingLoss	NAVI	0.00	0.02	0.11	5.13%
		ScanNet	0.04	0.26	1.43	10.97%
5	Matchability Predictor (A)	NAVI	0.00	0.02	0.17	5.55%
		ScanNet	0.00	0.00	0.20	5.95%
6	Matching Head (B)	NAVI	0.00	0.00	0.00	0.00%
		ScanNet	0.00	0.00	0.00	0.00%

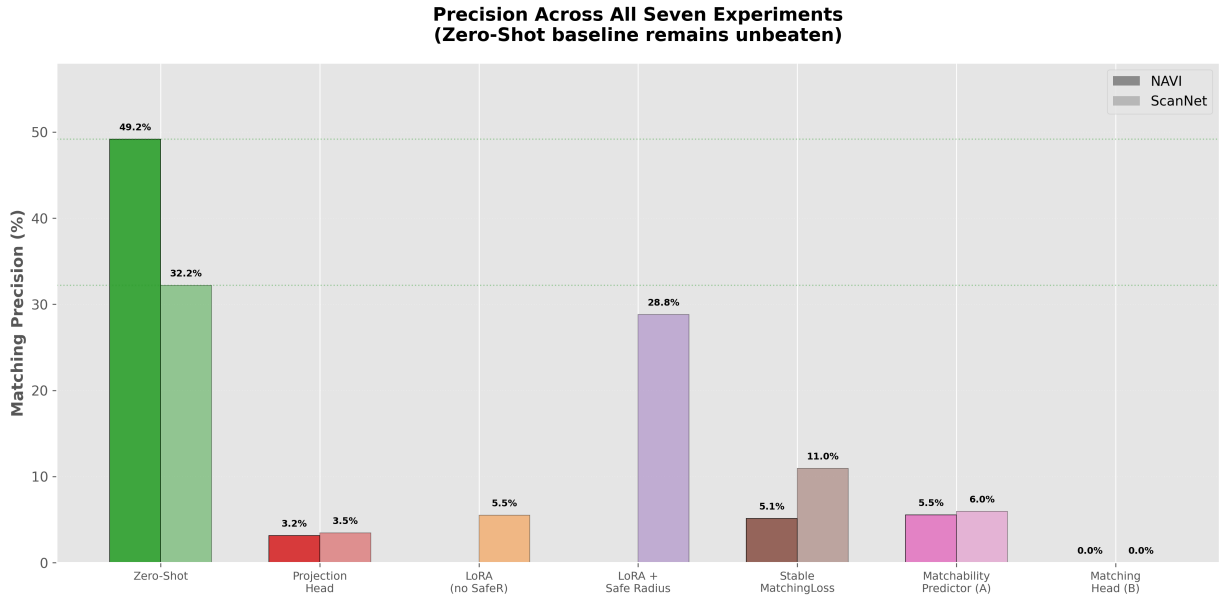

Figure 2: Matching Precision across all seven methods. Zero-shot baseline (green) is never surpassed. LoRA + Safe Radius (purple) is the single result approaching baseline.

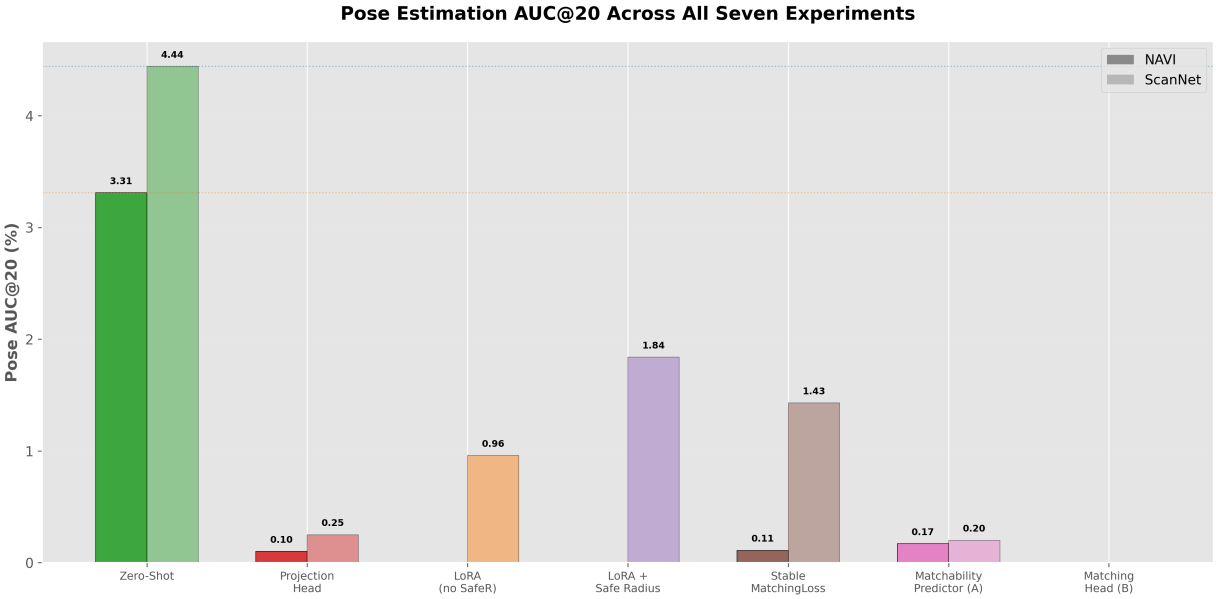

Figure 3: Pose AUC@20 across all methods. Every fine-tuning run degrades pose estimation quality.

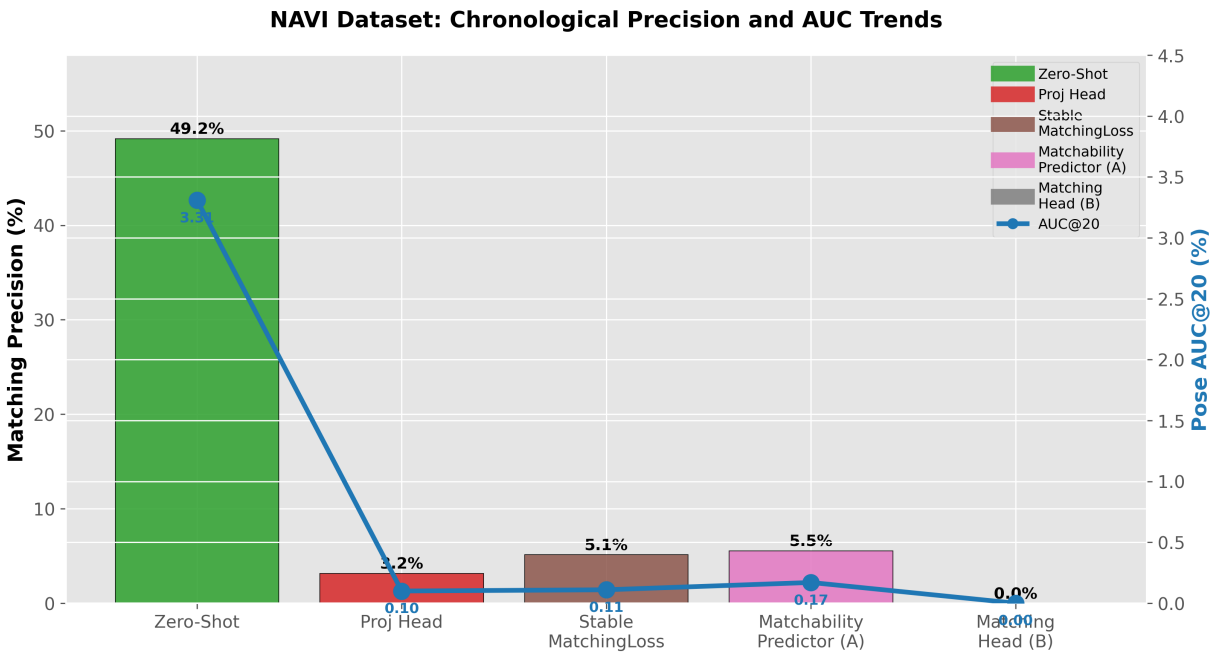

Figure 4: NAVI dataset: Precision and AUC@20 in chronological order. Each fine-tuning attempt produces a cliff-like drop.

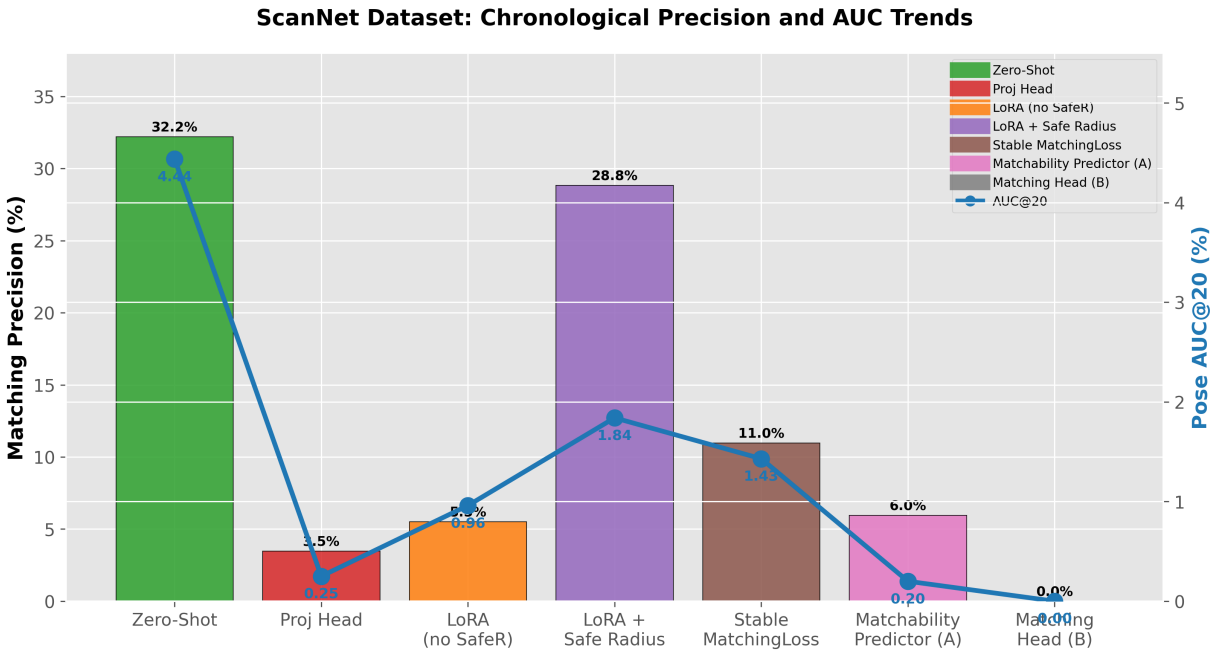

Figure 5: ScanNet dataset: chronological trajectory. LoRA + Safe Radius is the only upward deflection.

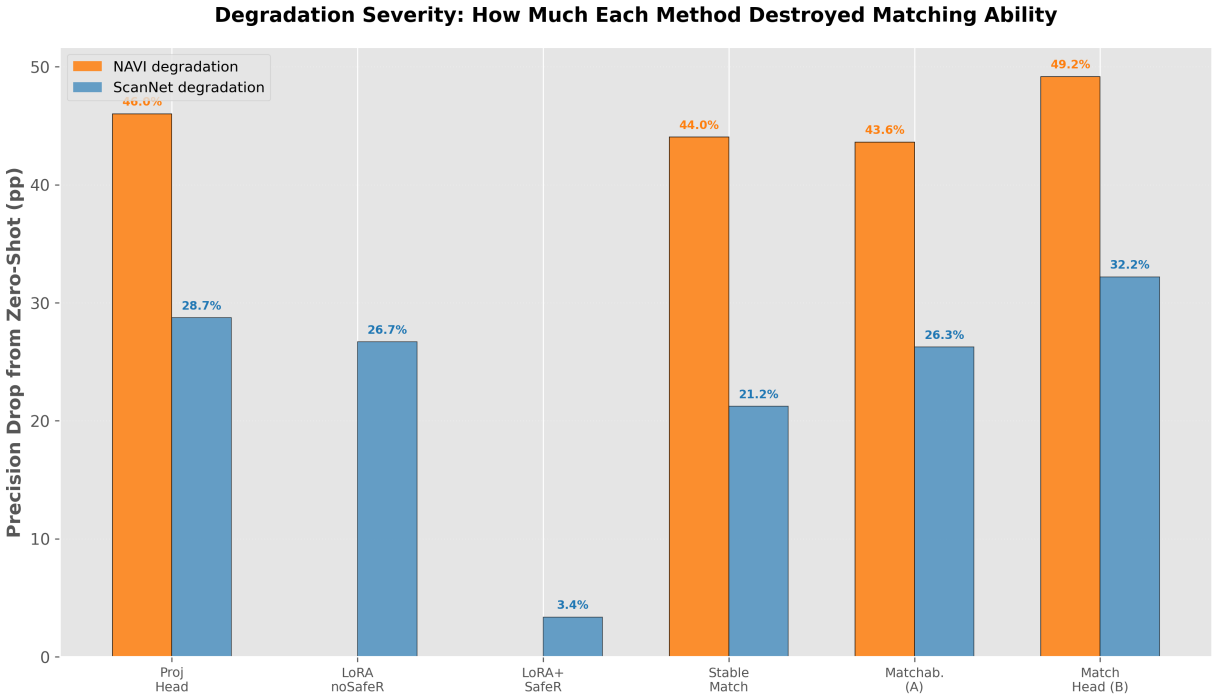

Figure 6: Precision degradation from zero-shot (percentage points). More sophisticated methods produce more severe degradation.

Table 4: Semantic vs. geometric features.

Property	Semantic (DINOv3)	Geometric (matching)
Similarity	"doorknob ≈ doorknob"	"*this* corner ≠ *that*"
Distribution	Smooth semantic regions	Sparse, distinct keypoints
Cross-instance	Generalize across instances	Discriminate within instance
Viewpoint	Viewpoint-invariant	Viewpoint-covariant
Ideal space	Continuous manifold	Discrete clusters

Three Paradigms, Seven Experiments, Zero Successes

Paradigm	Method	DINO Features	Best Precision	Verdict
Modify Features (Sec 3-5)	Projection Head LoRA + InfoNCE LoRA + Safe Radius	Modified (damaged)	3.5% 5.5% 28.8%	Catastrophic collapse Partial recovery
Filter Patches (Sec 6)	Matchability Predictor (A)	Frozen (pristine)	5-6%	Lacks context; filters good patches
Learn to Match (Sec 7)	Attention Matching Head (B)	Frozen (pristine)	0.00%	Zero geometric distinctiveness

Figure 7: Summary across all three paradigms. No approach improved upon zero-shot.

5 Root Cause Analysis

Seven systematic experiments converge on one conclusion: the failure is not engineering but fundamental. This section peels back seven layers of causation.

5.1 Layer 1: Semantic Features ≠ Geometric Features

DINOv3 learns **semantic features**: same-category objects are similar, different-category objects are dissimilar. Matching requires **geometric features**: each specific 3D point must be *unique enough* to distinguish it from all others, even within the same semantic category.

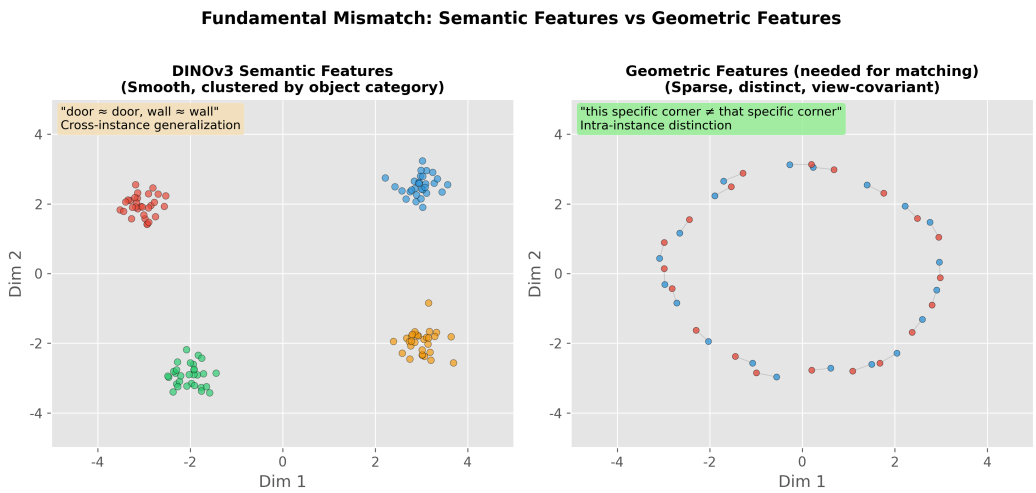

Figure 8: Left: DINOv3's semantic space—clustered by category, smooth. Right: Geometric space—sparse, distinctive, view-covariant.

5.2 Layer 2: Contrastive Learning Destroys Semantic Smoothness

DINOv3's zero-shot matching works *because of* semantic smoothness: the same object from different views lands nearby in feature space, where nearest-neighbor search finds correct correspondences. Contrastive fine-tuning destroys this: "pull positives" forces specific points to jump together; "push negatives"

Table 5: Pre-training vs. fine-tuning data scale.

Training stage	Data volume	Fraction
DINOv3 pre-training	~170M images	100%
Fine-tuning (NAVI)	5,596 pairs	~0.0003%
Fine-tuning (ScanNet)	1,500 pairs	~0.0001%

tears adjacent points apart. The result: a continuous manifold shredded into discrete clusters. Cross-view generalization is lost.

Paradox: the capability you are trying to improve depends on the very property you are destroying.

5.3 Layer 3: "Pulling Positives Together" Is Itself Wrong

In dense matching, a "positive pair"—two patches projecting to the same 3D point from different views—**should not have identical features**. Illumination differs across views; occlusion states differ; perspective distortion means the 16×16 patch covers slightly different surface areas; attention context differs across images. DINO features naturally encode these viewpoint differences—and this is a *strength*, because viewpoint difference is a critical signal for pose estimation. **"Pulling positives together" trains the model to discard the very signal needed for pose estimation.**

5.4 Layer 4: Three Non-Differentiable Gaps

$$\text{Cosine sim.} \xrightarrow{\text{non-diff.}} \text{MNN} \xrightarrow{\text{non-diff.}} \text{USAC + E} \xrightarrow{\text{non-diff.}} \text{Pose error} \tag{7}$$

The model never sees how improving cosine similarity affects the final pose. "Better features" from the training perspective can easily produce "worse poses" from the evaluation perspective.

5.5 Layer 5: The Data Chasm

Every gradient step overwrites vast pre-trained knowledge with extremely limited information. This is not fine-tuning; it is *micro-vandalism*.

5.6 Layer 6: The Information-Theoretic Ceiling

This is the deepest finding, crystallized from seven experiments:

> **DINOv3's training objective explicitly minimizes the mutual information between viewpoint variation and semantic features.**
>
> $I(\textbf{geometric distinctiveness}; \textbf{semantic feature}) \approx 0$
>
> **Geometric distinctiveness was eliminated as "noise" during pre-training. This is not a signal awaiting a better extraction method—it is information that has been irreversibly deleted.**

DINO's self-supervised learning requires "same object from different viewpoints → same feature." Viewpoint information—including geometric distinctiveness—is explicitly suppressed. Zero-shot matching achieves Precision 49%/32% only because this suppression is imperfect: residual "edge traces" of geometric signal survive. Those traces constitute the **performance ceiling**.

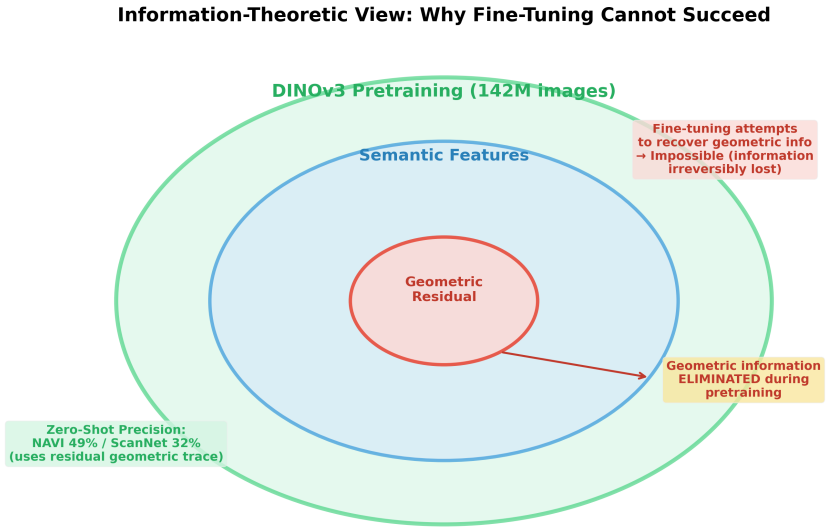

Figure 9: Information-theoretic view. Geometric information (red) was eliminated during pre-training. No fine-tuning can recover permanently deleted information.

Analogy. Consider an aggressively JPEG-compressed image. Post-processing (sharpening, denoising) can improve perceived quality but cannot recover quantized high-frequency details. DINOv3 features are the compressed image; geometric distinctiveness is the lost high-frequency detail; zero-shot matching uses the surviving low-frequency signal. All fine-tuning attempts are efforts to reconstruct high-frequency detail from low-frequency signal—information-theoretically impossible.

5.7 Layer 7: Sophistication ∝ Degradation

A deeply counter-intuitive finding: **more sophisticated methods produce more complete degradation**. Safe Radius (a 5-line mask) → 28.83% Precision. StableMatchingLoss (four components) → 10.97%. Matching Head (Self/Cross-Attn + Dual-Softmax) → 0.00%.

This is not coincidence. Sophisticated methods make stronger assumptions about DINO's feature space (e.g., "attention can distinguish semantically similar but geometrically distinct patches"). Under $I(\text{geometry} \mid \text{semantics}) \approx 0$, every such assumption is false. The sophisticated design becomes a precision destruction mechanism.

6 Conclusion

6.1 Feasibility Verdict

Fine-tuning DINOv3's backbone with contrastive learning to improve feature-matching-based camera pose estimation is methodologically infeasible.

This is not "we haven't found the right method yet." It is "the right method is to not do this."

6.2 Why Zero-Shot Is the Ceiling

DINOv3's zero-shot features work for matching (Precision 32–49%) precisely because of their **semantic smoothness**. Every form of contrastive fine-tuning destroys this smoothness. The loss function demands that the continuous semantic manifold be torn into discrete correspondence clusters. Once smoothness is destroyed, cross-view generalization collapses. More fundamentally: DINOv3's pre-training irreversibly eliminated geometric distinctiveness as noise. Fine-tuning can only operate on information present in the feature space—and that information no longer exists.

6.3 What Would Work Instead

1. **Better post-processing on frozen features.** SNN ratio test [12], improved keypoint selection, RANSAC parameter tuning—zero fine-tuning, direct zero-shot boost.

2. **End-to-end differentiable matching with sufficient data.** LoFTR [13] and RoMa [14] optimize geometric accuracy directly but require orders of magnitude more training pairs.
3. **Purpose-built geometric feature extractors.** SuperPoint [15], DISK, ALIKE are designed specifically for matching, unlike general-purpose semantic models.

6.4 Academic Significance

Understanding why something fails is often more valuable than demonstrating success: (1) A success tells you one path works; a failure analysis tells you why hundreds do not. (2) Our failure is not random but *necessary*—we eliminated every engineering variable and converged on a methodological root cause. (3) This conclusion guides future researchers: knowing what *not* to do is as valuable as knowing what to do.

6.5 Methodological Contributions

This project demonstrates a complete scientific workflow: baseline establishment, hypothesis-driven experimentation, mathematical diagnosis (proving mode collapse via $\ln(129)$), independent algorithmic innovation (Safe Radius), systematic ablation of all engineering variables, and deep methodological reflection.

Seven experiments, seven layers of understanding,
converging on a first-principles insight.

Proving the infeasibility of a direction holds
equal academic value to finding a viable one.

"What *not* to do" is often more important than "what to do."

A Complete Results Table

Complete Experimental Results: Seven Attempts Across Two Datasets

#	Method	Dataset	AUC@5	AUC@10	AUC@20	Precision	vs ZS
0	Zero-Shot (baseline)	NAVI	0.24	1.07	3.31	49.17%	—
0	Zero-Shot (baseline)	ScanNet	0.23	1.18	4.44	32.20%	—
1	Projection Head + InfoNCE	NAVI	0.00	0.03	0.10	3.16%	↓93.6%
1	Projection Head + InfoNCE	ScanNet	0.00	0.07	0.25	3.47%	↓89.2%
2	LoRA + InfoNCE (no SafeR)	ScanNet	0.00	0.09	0.96	5.51%	↓82.9%
3	LoRA + Safe Radius (bs=1)	ScanNet	0.00	0.26	1.84	28.83%	↓10.5%
4	LoRA + StableMatchingLoss	NAVI	0.00	0.02	0.11	5.13%	↓89.6%
4	LoRA + StableMatchingLoss	ScanNet	0.04	0.26	1.43	10.97%	↓65.9%
5	Matchability Predictor (A)	NAVI	0.00	0.02	0.17	5.55%	↓88.7%
5	Matchability Predictor (A)	ScanNet	0.00	0.00	0.20	5.95%	↓81.5%
6	Matching Head (B)	NAVI	0.00	0.00	0.00	0.00%	↓100%
6	Matching Head (B)	ScanNet	0.00	0.00	0.00	0.00%	↓100%

Figure 10: All experimental results. Green = zero-shot baseline, red = complete degradation (0.00%), yellow = partial preservation (28.83%).

References

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